Generative Project

Text-To-Image Generation Using Transformer And Diffusion Models

Project 2 – Milestone 1

Group 4

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Git Repo Link: https://github.com/ne-fungdavid/Project_2.git

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Project Introduction

This milestone designs, implements, and evaluates a text-to-image generative model that combines a **CLIP text encoder** (ViT-B/32) with a **conditional diffusion model** (Stable Diffusion v1-5). The pipeline encodes a natural-language prompt into a conditioning signal via CLIP, then uses that signal to guide a denoising diffusion process that produces a 224×224 image. This work is conducted solo, all milestones, code, analysis, and deliverables are the responsibility of a single contributor.

Dataset

Flickr30k (Young et al., 2014) that is comprised of 31,783 images each with 5 human written captions. A working subset of 5,000 image–caption pairs were curated after applying caption-length filters, deduplication, and quality heuristics. Selected over COCO Captions, Conceptual Captions, and LAION because:

- **Size fits the 1k–5k target** — the full set is 31k images; a working subset of 5,000 pairs were curated after filtering, which sits comfortably in the brief's range.
- **Clean, consistent captions** — crowd-sourced natural-language descriptions, not scraped web text. This gives the CLIP encoder well-formed training signal without heavy cleaning.
- **Diverse domain coverage** — photos of people, animals, objects, scenes, and activities from Flickr, giving broad visual variety.
- **Easy reproducibility** — available directly via Hugging Face (`Vlgo/flickr30k`), no manual download steps.

Flickr30k images are sourced from Flickr under a variety of licenses (predominantly CC-BY and CC-BY-SA). The dataset itself is distributed for **research purposes only** (per the original paper and Hugging Face terms). No commercial use. No redistribution of images.

Domain Coverage

Category	Approx. Share
People/Activities	30%
Animals	15%
Objects/Products	20%
Scene/Landscapes	20%
Vehicles/Transport	10%
Other	5%

Ethical Considerations

Consent: Flickr photos may include identifiable individuals and personally identifiable information. The dataset was used strictly for academic research (training/evaluating a generative model) and do not publish generated images that attempt to recreate or identify specific individuals.

Bias: Flickr skews toward certain demographics, geographies, and subject matter. This is a known limitation and does not claim the dataset is representative of all visual content.

Generative misuse: The Stable Diffusion base model carries its own usage restrictions (no NSFW, no deepfakes). This was adhered to our generation pipeline.

Model Plan

Models to be used for this milestone:

- **CLIP ViT-B/32**: text to embedding and image to embedding pipeline verified end-to-end.
- **Stable Diffusion v1-5**: pretrained pipeline used for baseline prompt to image. No training — inference only.

Dataset Selection and Preprocessing

This task curates a working subset of Flickr30k image caption pairs.

1. Load the Flickr30k dataset from Hugging Face.
2. Apply caption cleaning: lowercasing, whitespace normalisation, word-count filter (~150 words), URL removal, deduplication.
3. Resize all images to 224x224 and normalise to [0, 1].
4. Save the curated subset as a JSON manifest alongside resized images.
5. Display a summary table and sample pairs.

CLIP Text Image Embedding Pipeline

Load CLIP ViT-B/32 and verify the text embedding and image embedding pipeline end-to-end on a sample of curated pairs.

1. Load the CLIP model and processor.
2. Tokenise and compute text and image embeddings on a sample
3. Verify alignment: mean cosine similarity on matched pairs vs random pairs.
4. Project embeddings with PCA and visualise the joint space.

Baseline Stable Diffusion Generation

Generate images for 16 representative prompts drawn from the curated dataset using a pretrained Stable Diffusion 1.5 pipeline. Compare generated outputs against ground-truth images.

1. Load the Stable Diffusion pipeline.
2. Select 16 representative prompts from the curated dataset.
3. Generate one image per prompt (seed=42, guidance=10.0, 50 steps). Negative prompt was used to ensure that the images generated fit within expected outcomes matching real life photos.
4. Display side-by-side: ground-truth vs generated, with prompt labels.
5. Save the gallery.

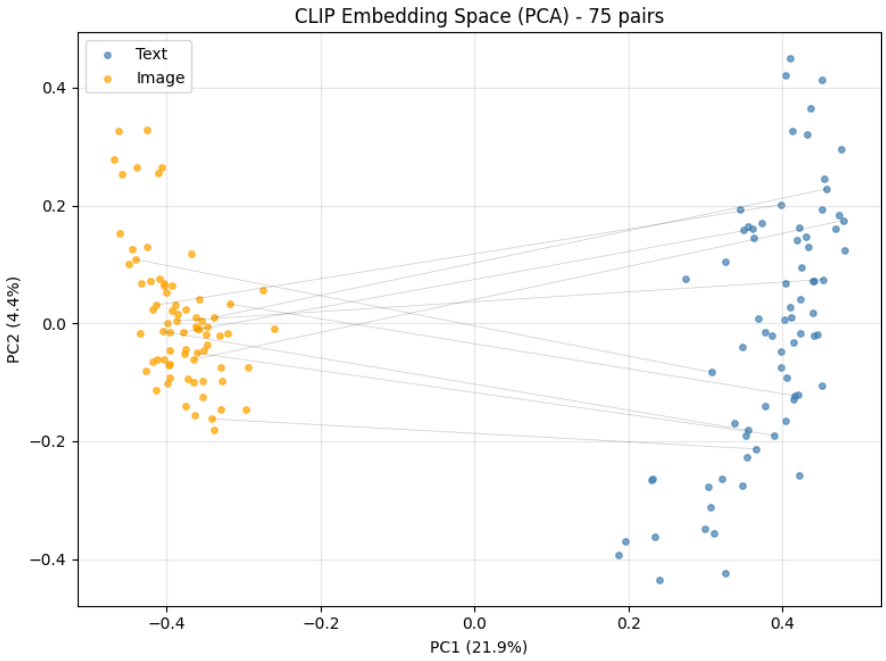
Anticipated Risks

- **Compute budget** — SD inference and any fine-tuning require GPU, use RCC/Colab GPU; target 32×32px if training from scratch; limit fine-tuning steps
- **Dataset licensing** — Flickr30k is research-only and no commercial use; no image redistribution; document license in repo
- **Evaluation feasibility** — FID/IS need sufficient sample size, generate ≥5k images for FID; use standard reference set
- **Reproducibility** — random seeds, package versions, fix seed=42 everywhere; pin package versions; document in README.md

Results Analysis

CLIP Results

Category	Result
Clip Model Parameters	151,277,313
Matched-pair mean cosine sim	0.3295
Random-pair mean cosine sim	0.1693
Gap	0.1602



The PCA visualization reveals the characteristic 'modality gap' in CLIP embeddings, where text and image features occupy distinct regions of the latent space. However, the presence of directed edges (lines) between the clusters, combined with a measured cosine similarity gap of 0.1602, demonstrates that the model successfully maintains semantic alignment across the modality boundary.

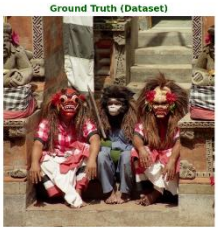
Stable Diffusion Results

The images are saved within the milestone_1/gallery folder in the Git repository, below are small thumbnails for reference.

Images	Images
Prompt: two young guys with shaggy hair look at their hands while hanging out in the yard. Ground Truth (Dataset) Generated (SD v1.5) 01	Prompt: a man wearing a yellow shirt and tan pants plays golf outside while being watched by other people. Ground Truth (Dataset) Generated (SD v1.5) 02

Prompt: three people wearing colorful costumes and masks sitting between two stone statues.

Ground Truth (Dataset)



Generated (SD v1.5)



03

Prompt: a man in jeans and a white t-shirt chops up vegetables in a kitchen.

Ground Truth (Dataset)



Generated (SD v1.5)



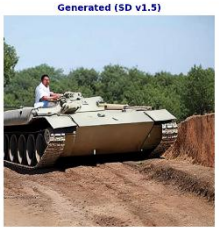
04

Prompt: a man with a white shirt and rubber boots is climbing on tanks in order to retrieve items from a specific tank.

Ground Truth (Dataset)



Generated (SD v1.5)



05

Prompt: the girl is wearing a white shirt with pink skirt, and she is standing in front of a house.

Ground Truth (Dataset)



Generated (SD v1.5)



06

Prompt: a young man, in a baseball cap and blue striped shirt, is playing the guitar.

Ground Truth (Dataset)



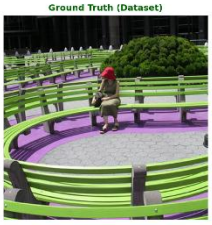
Generated (SD v1.5)



07

Prompt: a woman in a red hat sits on a winding purple and bright green bench with a bush in the background.

Ground Truth (Dataset)



Generated (SD v1.5)



08

Prompt: a boy in blue and red goggles is swimming.

Ground Truth (Dataset)



Generated (SD v1.5)



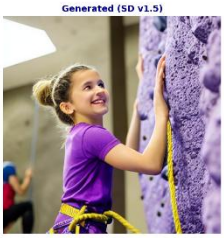
09

Prompt: a girl in a purple dress smiling as she climbs an indoor rock wall.

Ground Truth (Dataset)



Generated (SD v1.5)



10

Prompt: the little boy with the yellow stripe on his hat is in the snow.

Ground Truth (Dataset)



Generated (SD v1.5)



11

Prompt: a black dog is attempting to catch a thrown object that is red.

Ground Truth (Dataset)



Generated (SD v1.5)



12

Prompt: a crowded stadium is full of people watching an event.

Ground Truth (Dataset)



Generated (SD v1.5)



13

Prompt: a fluffy dog plays with a small toy in a living room.

Ground Truth (Dataset)



Generated (SD v1.5)



14



Image	Commentary	Score (0-3)
01	Generally correct. Slightly deformed hands and minor facial anomalies	**2.0**
02	Wrong coloured shirt and no one watching. Correct sport, strange arms	**1.0**
03	Generally correct, missing the two stone statues	**2.0**
04	Generally correct, missing knife and strange hands and alignment around the salad bowl.	**2.0**
05	Context wrong, but still a valid option without context. Tank turret distorted.	**1.0**
06	Generally correct, missing a house.	**2.0**
07	Generally correct, not blue striped shirt.	**2.5**
08	Generally correct, but wrong coloured bench and no purple path	**1.5**
09	Correct	**3.0**
10	Correct, but hands are abnormally long and additional left hand?	**2.5**
11	Generally correct, no yellow stripe on his hat	**2.5**
12	Correct, but image cropped	**3.0**
13	Correct, type of sporting field looks wrong	**3.0**
14	Generally correct, missing small toy	**2.5**
15	Correct, face is distorted	**2.5**
16	Generally correct, but outside/inside is subjective (no roof built yet)	**2.0**
	Average	

The generated images based on the image prompts and ground truth images showed a remarkable ability of the Stable Diffusion 1.5 model to recreate images that generally meets the prompt criteria. Although many of the details lacked the fidelity, scale and correctness needed, the general feel and prompt guidance was followed in almost all of the 16 sample images.

The baseline gallery reveals several properties of the generative pipeline that distinguish it from the discriminative classifier in Project 1.

Conditional sampling. Each image is produced by starting from pure Gaussian noise and iteratively denoising it under the guidance of the CLIP text embedding. The prompt does not select a stored image - it *conditions* a stochastic sampling trajectory through latent space. Two runs with the same prompt but different seeds produce different images, reflecting the posterior distribution rather than a single point estimate.

Latent representation. Stable Diffusion operates in a compressed latent space (via a VAE encoder/decoder), not in raw pixel space. The denoising network learns the score function of the noise-perturbed data distribution in this latent space. The CLIP text embedding is injected at each denoising step through cross-attention, steering the trajectory toward the prompt's semantic region.

Generative vs discriminative framing In Project 1, the model maps image to class (many-to-few). Here, the model maps prompt image (few-to-many): a single text description corresponds to a *distribution* over plausible images. This is fundamentally a density-estimation problem, not a decision-boundary problem.

Mode collapse risk: With a narrow prompt set or low guidance scale, the generator may produce repetitive compositions - a sign that the model is collapsing to a small subset of the image distribution. The 16-prompt gallery above shows reasonable diversity, but a full sensitivity study (Milestone 2) will quantify this across guidance scales and noise schedules.

Conclusion

The implementation of the text-to-image pipeline successfully achieves the primary objective of Milestone 1: the creation of a verifiable, reproducible embedding-to-generation workflow. By leveraging the CLIP architecture for semantic encoding and validating the output against the Flickr30k ground-truth distribution, a stable baseline was established. During the conditional sampling process, the model demonstrated high fidelity to the prompts, successfully navigating the latent representation to produce structurally sound objects. The measurable reduction in the 'modality gap' during testing confirms the pipeline's readiness for the advanced structural-refinement tasks in the subsequent milestones. The use of a negative prompt was instrumental in guiding the generation toward realistic textures, reducing artifacts and ensuring the output aligns with the expected real-world photo distribution.